PROGYA PAROMITA BISWAS

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EDUCATION

Vellore Institute of Technology 2023 – 2027
B.Tech in Computer Science and Engineering; CGPA: 8.5/10
Dr. Graham's Homes 2010 – 2023
Secondary (ICSE, 2020–2021) | Higher Secondary (ISC, 2022–2023)

RESEARCH EXPERIENCE

Real-Time Distance Estimation using Monocular Vision Jan 2026 – Present
- Developed a system that estimates object distance in real time from a single RGB camera, implementing monocular depth and distance estimation algorithms.
- Optimized the inference pipeline for real-time performance; paper submitted to IEEE ICAITPR 2026 and currently under review. Potential applications include autonomous systems and assistive technology for visually impaired individuals.

Energy-Aware Parallel Computation of SHAP Explanations Aug 2026 - Present
- Benchmarked Serial, Multiprocessing, Threading, Joblib, and Ray strategies for TreeSHAP and KernelSHAP, completing 972 runs across 324 configurations with CPU time and energy measured using CodeCarbon.
- Analyzed workload-dependent time-energy trade-offs across Random Forest, XGBoost, SVM-RBF, and Logistic Regression to generate time-optimal, energy-optimal, and balanced CPU scheduling recommendations.

Reinforcement Learning-Based Multi-Robot Navigation Feb 2026 – Apr 2026
Research Paper, Robotics Lab, VIT Vellore
- Implemented autonomous mobile robot navigation using DQN and PPO reinforcement learning.
- Integrated a Waveshare robot chassis, Jetson Nano, RPLiDAR, and camera modules with ROS2/Gazebo for autonomous navigation; designed the sensor fusion and real-time communication architecture.

TECHNICAL PROJECTS

Early Disease Detection from Nail Images May 2025 – Present
- Developed and evaluated multiple deep learning architectures, including Xception and VGG16, for nail-image disease classification using preprocessing, augmentation, and transfer learning.
- Achieved 89% validation accuracy with Xception for automated early-stage disease detection, with further architectures being evaluated for improved classification performance.

HorizonEye — Vision-Based Horizon Detection & Height Estimation Aug 2026 – Present
- Developed a camera-based flight perception system using sky-ground segmentation, Otsu thresholding, and horizon line fitting to estimate aircraft roll and horizon dip angle.
- Implemented camera and spherical-Earth geometry to estimate flight altitude using Python, OpenCV, and NumPy.

Sound Energy Harvesting System using Piezo Sensors Jan 2026 – Feb 2026
- Designed a piezoelectric system to convert ambient sound vibrations into usable electrical energy for low-power applications.
- Implemented energy storage and power optimization circuits to improve energy utilization and system efficiency.

LEADERSHIP

Team Lead, Sound Energy Harvesting System using Piezo Sensors Jan 2026 – Feb 2026
- Conceived the idea and led a 2-member team to develop a piezoelectric energy harvesting system that converts ambient sound vibrations into usable electrical energy for low-power applications.

Team Lead, Reinforcement Learning-Based Robot Navigation, *Robotics Lab, VIT Vellore* Feb 2026 – Apr 2026
- Led a 3-member research team, under faculty guidance, to develop an RL-based multi-robot navigation system.

Team Lead, CookBook Web Application Feb 2026 – Apr 2025
- Led a 4-member team to design and build a responsive recipe-sharing web application in ReactJS, with interactive features for recipe sharing and community engagement, integrating frontend components with backend services.

TECHNICAL SKILLS

Languages: C++, C, HTML, CSS, Python, Java, MySQL, Oracle Database
Intelligence: Machine Learning, Deep Learning, Computer Vision, Image Processing, TensorFlow, Keras
Robotics: ROS2, Gazebo, Sensor Integration, Real-time Systems
Tools: Git, GitHub, Linux (Ubuntu), VS Code, MATLAB, Jupyter Notebook, Google Colab, Cisco Packet Tracer, Keil, Arduino IDE, ModelSim, StarUML